# AI-Driven Platform Guiding Paddy Farmers Across the Full Cultivation Lifecycle

Project Id: 25-26J-242

**Individual Report**

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BSc (Hons) in Information Technology Specializing in Information Technology

**Department of Information Technology**

Sri Lanka Institute of Information Technology

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## Declration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name | Student Id | Signature |
|---|---|---|
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The above candidates are carrying out research for the undergraduate Dissertation under my Supervision.

..... 29/08/2025

Signature of the supervisor: Date

## ABSTRACT

Agriculture is the backbone of the Sri Lankan economy with paddy field cultivation being the most widely practical farming activity in the around the Sri Lanka But not having technology for paddy field cultivation, So it is most sadness thing about the Sri Lankan technology world. All farmers and new wanted to be farmer with having not expiring knowledge of the paddy yield are conduct the low property from the paddy yield. Our very obedient object also get more yield and farmers having new life with technology changes and adopt the technology world with using our guidance system. There is some despicably in the sector's the signification, paddy framers are faced numerous challenges that have threaten by the productivity, sustainability and livelihood. These issues have largely due to the outdated practices and not having including world technology, limited access to the real time data and lack of the advanced technology support for farmers. Traditional farming methods are reliant on human intuition, irregular weather patterns, and generic agricultural calendars like are no longer sufficient in the face of increasing climate variability, soil degradation, and pest attackers. Many small farmers have lack of access to the modern tools and actionable insights, and easy farming technology used for other countries which leads to inconsistent yields, inefficient resource use, and economic hardship. So having all solution in one system is really reliable for agricultural field future knowledge and gain new updated all over the world matching system.

In this chapters all over I would like explain my individual components of most important and all yield property are based on Pest Attack forecasting component. Pest infestations is often detected too late then full of all harvest effect and get low harvest because of this detective is to doing the too late. Can wipe out entire fields, as most farmers have no reliable knowledge about this pest effected time to predict or prevent outbreaks. While having very advanced varieties offer resistance and having new pests and disease can metalize and over certain solutions can lead the clearly resistance develop pests. Pest really hard to manage if we don't do the right method for solution for type of pest behaviors and also without having grown up knowledge of the pest specifies. This system used historical weather data, crop data, pest infection data records for upcoming pests attack by notify altering with 7 day forecast, famer in advanced it helps farmers to the reduce pesticide overcome with overdue crop damages and profit loss. Also having the YOLO modeling-based pest detection with identify pest with image inputting from field is get more identify the pest detection with high level model training enhanced the pest management with recommended fertilizers, with pest heat maps also. It must be learned from the past outbreaks data predictive according to the pest movement in crops ages and behaviorists and alert's farmers before it's too late to the do the treatment for the crops. In this case framers can know about what kind of pest attacks coming for what the reasons like weather situation changes and ages of the paddy crop can easily suffer from attacks and also paddy seed types. Humidity level, soil conditions, pest data from past as multiple real- time data using multimodal input allows to the AI predict more accuracy by learning how to different environmental factors contribute to pest outbreak in specific locations and different stages of the paddy growth. This Depth is not offered in most Rule based or weather model. Most existing tools require manual data entry or inspections because this function empowers the farmer to act as real-time data collector and the model learns and improve with image classification of improving model making community- driven adaptive system.



**Keywords: Pest Management, forecasting, YOLO Model, crops, real-time data**

## ACKNOWLEDGEMENT

I want to thank everyone who helped me in completing this proposal successfully. Above all, I want to thank my supervisor, Mrs. Lokesha Weerasinghe, for her constant support, suggestions, and motivation towards the completion of this project. Her help made a lot of difference in the success of this proposal. Next, I would like to thank my co-supervisor, Ms. Shalini Ruupasinghe, for her assistance and help towards the completion of this project.My sincere thanks goes to the Department of Information Technology at SLIIT who provided us with all the knowledge and information needed to complete this system. In addition, I want to thank my fellow lecturers, friends, and colleagues who were there to provide assistance in terms of discussion, suggestions, and motivation. Lastly, but most importantly, I want to thank my parents for their continuous support towards the completion of this proposal.

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## LIST OF ABBREVIATIONS

LSTM – Long short Term Memory

DCNN – Deep Conventual Neural Network.

CNN - convolutional natural networks

# 1. INTRODUCTION

## 1.1 Background & literature review

Insect infestations remain the greatest challenge to paddy cultivation in Sri Lanka, causing severe losses in yield and financial hardships for the farmers annually. There life cycle is conduct with the climate changes, locations, diapauses like situations. So traditional approaches to forecasting such as seasonal calendars, field observations, and generalized warning remain incomplete, reactive, and non-responsive to the rapidly changing pest behaviors caused by climate variability. So this kind of traditional methods are not useful in the furthermore. This leads to delayed detection, overuse of pesticides, and low productivity. Current advances technology such as in Artificial Intelligence (AI), machine learning, and computer vision also now make it possible to build dynamic, data-driven pest forecasting systems that can analyze multiple environmental factors simultaneously Using real time data factors and reasons. AI models are also able to predict outbreaks more accurately earlier by integrating past data on pest outbreaks, weather, soil type, crop growth stages, and field images provided by farmers.

Having below kind of novelty features these components gave best of the distinguish it from the existing other than pest prediction tools:

1. **Localized of Sri Lanka-specific Modeling** – AI models designed on regional pest fauna, paddy crops, and monsoon-driven climate patterns, rather than generic global datasets.

2. **Multimodal Data Fusion** – Fusing meteorology data, soil parameters, historical pest data, and real-time farmer images to make accurate, location-specific predictions.

3. **Dynamic Learning** – The system continually updates predictions throughout the cultivation period based on real-time field data, accounting for unexpected climate fluctuations and pest behavior adjustments.

4. **Community-supported Data Collection** – Pest images and outbreak reports are submitted by farmers, gradually enhancing the accuracy of the system while fostering user participation.

5. **Stage-specific Actionable Recommendations** – Instead of simply predicting pest occurrence, the system provides targeted, crop-stage-based control measures, eliminating duplicative pesticide use and cost.

6. **Pest detection** – Detection pest is using YOLO identification with all of the field images and camera using inputting farmer

7. **Heat Mapping** – Map version of pest distribution in Sri Lanka based on dataset.

This synergism of local adaptation, constant learning, multimodal integration, and actionable guidance puts the system in a position to resolve Sri Lankan paddy farmer's actual-world pest management problems in a very unique manner from the technology world and having new technologies also.

## 1.2 Literature review

Pest attack forecasting has been a vital component of precision agriculture with tremendous potential in reducing yield loss, simplifying pesticide application, and enhancing environmental sustainability, particularly for weather-sensitive crops such as paddy. Traditional forecasting methods in Sri Lanka and the majority of developing nations rely mainly on fixed seasonal calendars, farmer experience, and manual scouting, which are inadequate in terms of providing early warnings or adapting to changing pest trends [1]. Meteorological conditions such as temperature, humidity, and rainfall have been discovered to greatly influence the dynamics of pest populations, triggering the application of time-series models such as Random Forests, as well as advanced deep learning methods such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks for temporal pattern recognition [2].

"Wang at.el [3] introduced architecture such as let Net and Alex net to classify various crop pests. Besides, computer vision techniques specifically Convolutional Neural Networks have been trained to identify pests and pest-induced crop damage from field images using large datasets, even though there is still a problem of data scarcity, class imbalance, and the need for pest species localization [4]. "Image segmentation and other approaches have been used to preprocess the images, after which the features of target images are extracted. Then machine learning methods, such as random forest, k nearest neighbors, support vector machine (SVM) , etc are used to perform pest identification." [5] [6] [7] [8]. "Although the approaches described above have produced good identification results, they have flaws such as insufficient extracted information, manual identification of critical features, and low feature extraction efficiency. Deep convolutional neural networks (DCNNS) have become an effective method to solve the problems mentioned above in recent year" [9].

"Furthermore, deep learning technology based on convolutional natural networks (CNN) has made significant progress in identifying objects, such as pests. Due to the poor identification performance of CNNs in the field, [10] proposed data augmentation approach based on CNNs". [11] Image background removal was performed during data acquisition using the Gaussian mixture model to aid the classification and identification of insects in the images." [12]

The methods are based on saliency map and applied it to DCNNs for pest identification performance for CNNs and found that the accuracy was better than that previous one. These hybrid approaches, often based on attention mechanisms or ensemble learning, outperform single-source models and can issue dynamic, location-specific predictions rather than static seasonal alerts. However, most existing systems are customized to commercial agriculture in developed nations and are not suited to smallholder farming contexts in Sri Lanka, where pest diversity, rice variety-specific vulnerabilities, monsoon-driven climatic uncertainty, and limited rural internet connectivity significantly affect outcomes [13] Most models also fail at prediction by not converting outputs into actionable, stage- specific pest control recommendations. Key gaps in the literature include the absence of Sri Lanka-specific pest forecasting models tuned on local historical outbreak data, the paucity of lightweight offline-enabled AI apps for low-connectivity

environments, and the absence of forecast integration with cost-sensitive intervention suggestions. The traditional monitoring methods obtain pest information by regular field surveys. Although field survey can obtain detailed pest information, it is time consuming, labor-intensive, and limited in large scale surveys. [14] So having below advanced novelty features is very usefully and user friendly [15].

“Cheng et.al [16] used a deep residual network to identify to the 10 different categories of crop pests in complex background. In comparing with SVM and back propagation, their model performs better and achieved and accuracy rate of 98.67%.”

“Temperature-based risk maps are widely used to represent the ability to develop and establish in a given geographical area [17]. They are of particular interest to policymakers because they can be used in integrated pest management to predict how the biodemographic characteristics of a given species would change in response to temperature variation over large spatial scales (regional, national, or continental).Some tool have been developed to obtain such risk maps and ae generally made available to end users either as R packages [18] or as decision support systems software such as Maxent [19], [20] ILCYM [21] or CLIMEX [22] [23]”.

“The mathematical expression to produce maps that help to visualize the variability of risk at reginal and continental scales is usually linked to a climate database to extract the values of the environment variables ,defined as input for the risk assessment [24] [25]”.

“Image based automatic pest identification system can monitor the pest type identification with pests’ problem solving methods can laborious manual identification and improve the reliability with many ways using replaced in thus providing the powerful tool with real-time pest information”[26].

Addressing these information and gaps using community-driven data curation, transfer learning for pest image classification, and climate-adaptive temporal modeling can enhance early detection accuracy and ease farmer adoption. In summary, the literature leads one to conclude that the best pest forecasting solutions
for paddy cultivation must be localized, data-intensive, and farmer-focused, marrying predictive accuracy with actionable advisories that are driven by socio-economic ground realities.

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### 1.3.Research gap

| Key Features | Research 1 | Research 2 | Research 3 | Propose Research |
|---|---|---|---|---|
| Uses multimodal data (weather + soil+ seed type + crop stage + pest) | Weather only ☒ | Weather only ☒ | Weather only ☒ | ☑ |
| Farmer Participation (Upload photos + Field data ) | ☑ | ☒ | ☒ | ☑ |
| Adaptive / Continuous learning AI | ☒ | ☒ | ☒ | ☑ |
| Computer Vision for pest Identification | ☒ | ☒ | ☒ | ☑ |
| Real-time forecasting Mobile /web Alerts | ☒ | ☒ | ☒ | ☑ |
| Pest detection with fertilizers recommendations | ☒ | ☒ | ☒ | ☑ |
| Heat Maps with pest Distribution | ☒ | ☒ | ☒ | ☑ |

| | | | | |
|---|---|---|---|---|
| Practical Farmer Decision support | ☒ | ☒ | ☒ | ☑ |

*Table 1.3.1 Research Gap*

## 1.4.Research problem

Pest attack forecasting is a critical component of modern agricultural systems for Sri Lankan paddy farmers, in which unexpected infestations of pests can devastate crops and livelihoods. The AI-Driven Integrated Agricultural Support Platform's Pest Attack Forecasting components aims to address this by applying state of the art machine learning techniques to predict pest infestations with high accuracy. By calculating past weather patterns, soil, and past infestation history, the system identifies patterns and trends reflecting possible infestations by pests.

As proactive as it can be, farmers can take countermeasures before infestations by pests happen and devastate crops in massive proportions, with minimum pesticide consumption and crop loss. The component is refreshed in real-time with reports of pest observations by farmers and hyper-local weather forecasts, soil type so its projections remain relevant throughout the season. Fixed rules or past data-based rule-based or seasonal forecasting systems, however, are based on pre-programmed rules or historical observations and fail to change according to changing environmental conditions or pest dynamics. For instance, it considers humidity levels, crop growth stages, and paddy seed varieties in order to forecast according to some context. Farmers are alerted by forecasting web notification in advance for having if near have pest attack, and they are able to apply proper interventions, Eg:- spraying with bio-pesticides or adjusting irrigation schedules like this. The incorporation of community-contributed data also enhances the precision of the system because farmers contribute images and field reports that enrich the AI models. Such co-operation not only increases prediction precision but also motivates farmers to maintain a knowledge-sharing culture.

Pest Detection with Fertilizer recommendations by the instants finding identify the pest to the , farmers can the receive recommendations and also it help for the have the recover pest attacking fast. Also have the farmers are questioned like how the maintain the fertilizers and how to face and buried the pest attacking not have furthermore in one system.

By overcoming the disadvantages of conventional pest control protocols, the Pest Attack Forecasting module is a reflection of how AI can transform agriculture into a more robust, sustainable, and productive sector. Its novelty lies in how it can combine multimodal data inputs, adaptive learning, and farmer engagement to deliver precise, actionable insights that guard crops and optimize resource use. In total, this innovation advances the overall mission of the platform is empowering Sri Lankan paddy farmers with data-driven instruments that enhance productivity, profitability, availability and climate resilience according to the various of paddy type.

## 1.5.Research Objectives

### 1.5.1.Main Objective

The primary objective of the Pest Attack Forecasting component is to establish an AI-driven predictive system for enhancing pest control for Sri Lankan paddy growers. According to past pest data, present environmental factors, and machine learning protocols, the system will issue early and accurate warnings of oncoming pest infestations. Pest Detection with fertilizer recommendations will more benefits. This early warning system tries to curtail crop loss, prevent excessive pesticide use, and maximize overall yield sustainability. In addition, the system introduces farmer-submitted data to make predictions more refined to the extent that the measures implemented are localized and also responsive to the stages of crop growth and territorial climatic patterns.

### 1.5.2. Specific Objectives

Timely correct prediction of Pest Outbreaks – Examining historical records of infestation alongside trend analysis of weather, crop type and soil in order to predict when infestations are to occur.

Alerts – Send SMS and mobile notifications web notification to farmers, hence equipping them with the capability to initiate prompt preventive measures on time.

Reduce Pesticide Reliance – Optimize pest management by recommending interventions only when necessary, lowering costs and environmental impact.

Adapt to Local field Conditions – Leverage farmer-submitted data Eg: pest photos, field observations to

regularly refine prediction models for local variations.

Pest detection with real time fertilizers recommendations.

Pest Heat map with localization pest distribution.

Enhance Farmer Decision-Making – Make actionable recommendations on pest-resistant crops and optimal treatment timing to assist in optimizing harvest quality and profitability accurately.

Enhance and improve the usability issues: Refine the sentiment analysis models frequently in response to user comments and changing linguistic trends. Create for farmers feedback loops so the system may learn from user interactions and gradually improve its replies. In order to ensure a flawless and easy to use user experience, address technical difficulties and constraints.

Through such objectives, the Pest Attack Forecasting system aims to transform traditional reactive pest control into a clever, Databased process that is more efficient in agriculture.

# 2. METHODOLOGY

## 2.1Understanding key pillars of the research domain

The pest attack forecasting system employs a well-organized, data-oriented methodology to forecast pest attacks and enable farmers in Sri Lanka to make proactive decisions regarding their paddy crops. The methodology employs machine learning, real-time data analytics, and community feedback to assure accuracy, flexibility, and scalability.
The process begins with such as data acquisition, retrieving pest history records, climatic trends temperature, humidity, rain, soil health data, and crop growth stages from agronomic databases and weather API. Preprocessing is performed on the data to address missing values, feature scaling, and remove noise. Supervised learning models, Random Forest, Gradient Boosting, or Long Short-Term Memory (LSTM) networks, are trained to recognize correlations between environmental factors and pest infestations. The machine learning model automatically updates its predictions constantly using real-time field data from farmers, including pest reports and crop health images shared via a mobile app or website. Hyperlocal weather forecasts from weather stations add more accuracy. For usability, the system offers self-contained notify and in-app alerts when it has the face the pest attack before it big to the full of field pest risk thresholds are exceeded, along with personalized countermeasures. Pest detection with YOLO V8 has classification with in many ways for the pest types. It has a feedback loop for farmers to submit outcomes, facilitating continuous model tuning through reinforcement learning. This methodology ensures correct prediction despite changing climate trends and pest patterns, ultimately forming the basis for sustainable and efficient pest control for paddy culture.

Using Software Development life cycle we can describe it simply.

1. Planning:– What are the things we need to achieve in doing pest attack forecasting research and how to get data for the do our project and also how to this affective in real time.

2. Analysis:- We need to analysis for our finding data and what kind of software using be okay for our project.

3. Design:- Design the AI models ML algorithms using the time series , weather data , pest data and also paddy data . And also define what are the things should be input data and output information data and also start design UIUX field in system.

*Figure : 2.1.1 Software development*

*Figure 2 Software Development Life Cycle*

2. Implementation:- Collected data using and trained model using like LSTM for time series forecasting .Build backend for API connections for forecasting alerts . YOLO based on Pest Detection with high accuracy when the detecting. Also develop mobile web-based system.

3. Testing and Integration:- In this phase we can ensure system works accurately before handover the client. We can used for this testing type like automation testing, performance testing, API testing, unit testing methods. After all testing we can have finally bugs free validate system.

4. Maintains:- Keep system update with real time and improve a ML model using updated data.

### 2.1.1 Deep Learning

Deep learning is an area of machine learning that utilizes artificial neural networks comprising several layers that are capable of learning from large volumes of data. It draws its inspiration from the workings of the human brain. It is capable of handling complicated inputs such as images, text, and voice without extracting features manually. In the system deep learning is used for the YOLO V8 for extract images with high quality of the detection for the pest identify the correctly with smoothly with high accuracy. Also Used a XG boost Random Forest Also for the pest forecasting classification. Deep learning has been applied in a number of areas, including computer vision, natural language processing, and autonomous driving.

### 2.1.2 YOLO V8 Classification pest Identifying

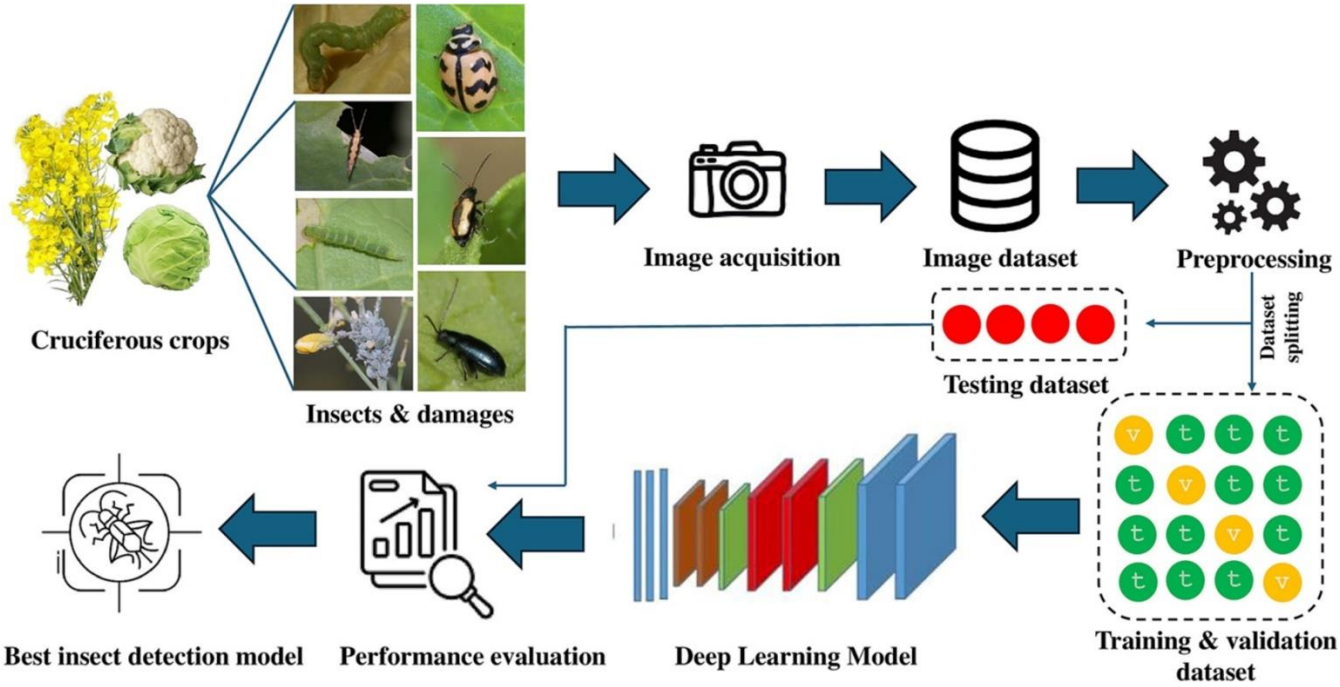



*Figure 2.1.2. 1Pest detection architecture*

*Source : https://www.sciencedirect.com/science/article/pii/S2772375524002685*

Once User upload the pest image then system was generate the image with the labeling and classifying with many ways . In the images when the processing detect the shapes, textures , patterns are identifying the unique fields.  Then based on accuracy system will evaluate the metrics and the save the best model applications. Then the training model working for the detect the real time Pests.

### 2.1.3 XG Boosting and Random Forest

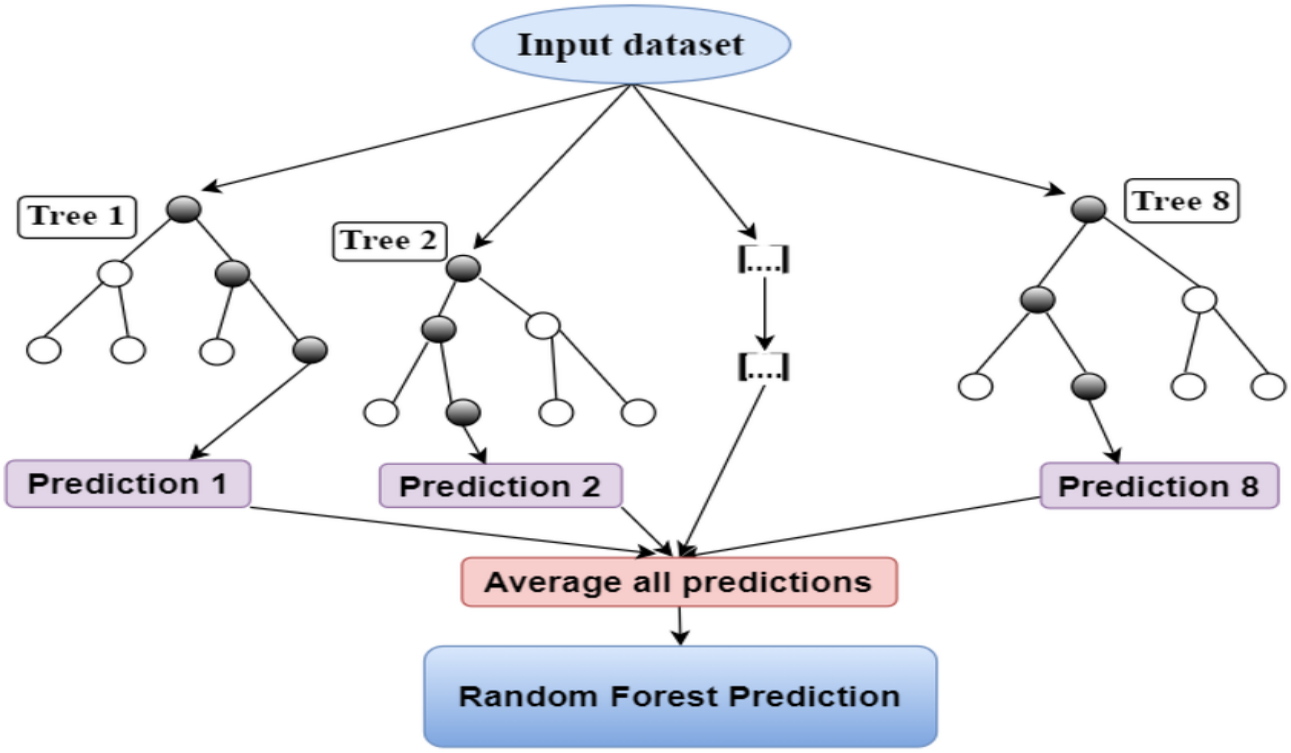



*Figure 2.1.3.1  Random Forest Classification*

*Source:* [https://www.researchgate.net/figure/Random-forest-regressions-model-architecture_fig4_382453093](https://www.researchgate.net/figure/Random-forest-regressions-model-architecture_fig4_382453093)

System is used the machine learning for the predict pest the outbreaks of the paddy fields. XG Boos t used for the identify the type of pest and Random forest classification is used for the make the predication based on the data set  . It predicts the severity level, Gradient Boosting estimates the percentage accuracy. System is used the weather, soil, and crop data for the with engineered featured along.

## 2.2Data Collection

Development depends on multimodal data such as weather patterns, soil health records, pest/weed incidence history, crop images, farmer feedback, and satellite data. Data on local market prices, transport costs, and farmer demographics are also vital for contextual relevance. Continuous collection and integration of farmer-uploaded data e.g., pest images, harvest logs will fuel active learning models, while GIS-based land profiling and real-time weather API input support precision agriculture. All data collection must comply with ethical standards, including farmer privacy and informed consent.

Pest Data:

*Figure 2.2.1 BPH Hopper*

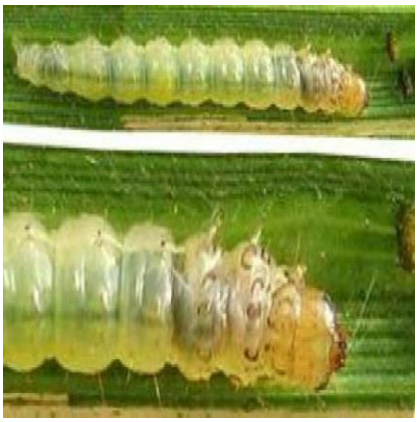


*Figure 2.2.2 Leaf folders*

*Figure 2.2.3 Rice Bugs*

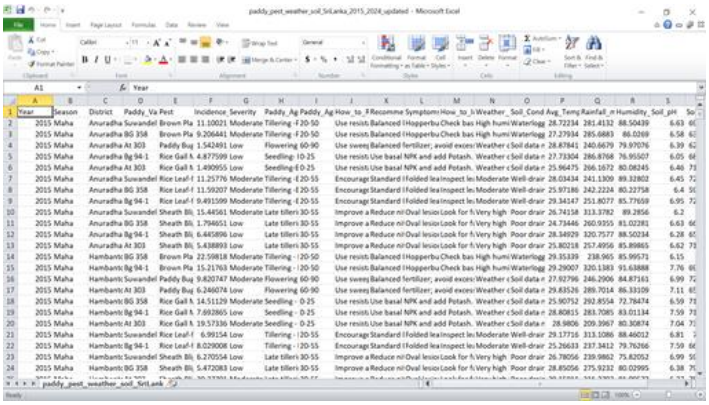


*Figure 2.2.4 Data Set*

## 2.3Summary of the Methodology

| Step | Stage | Description | Techniques/Tools Used |
|---|---|---|---|
| 1 | Data collection | Load the historical data | Pandas |
| 2 | Data Preprocessing | Clean and handle missing data | NumPy, Pandas |
| 3 | Feature Engineering | Create the meaningful features | python |
| 4 | Data encoding | Convert categorical variables | Data encoding |
| 5 | Data Splitting | Split data into the training data set | Train_test_spliit |
| 6 | Pest predict model | Predict the pest type using boosting algorithm | XG Boost classifier |
| 7 | Model Evaluation | Evaluate model using accuracy | Sklearn metrics |

*Table 2.3.1 summary of the methodology*

## 2.4 High-level Architectural Diagram

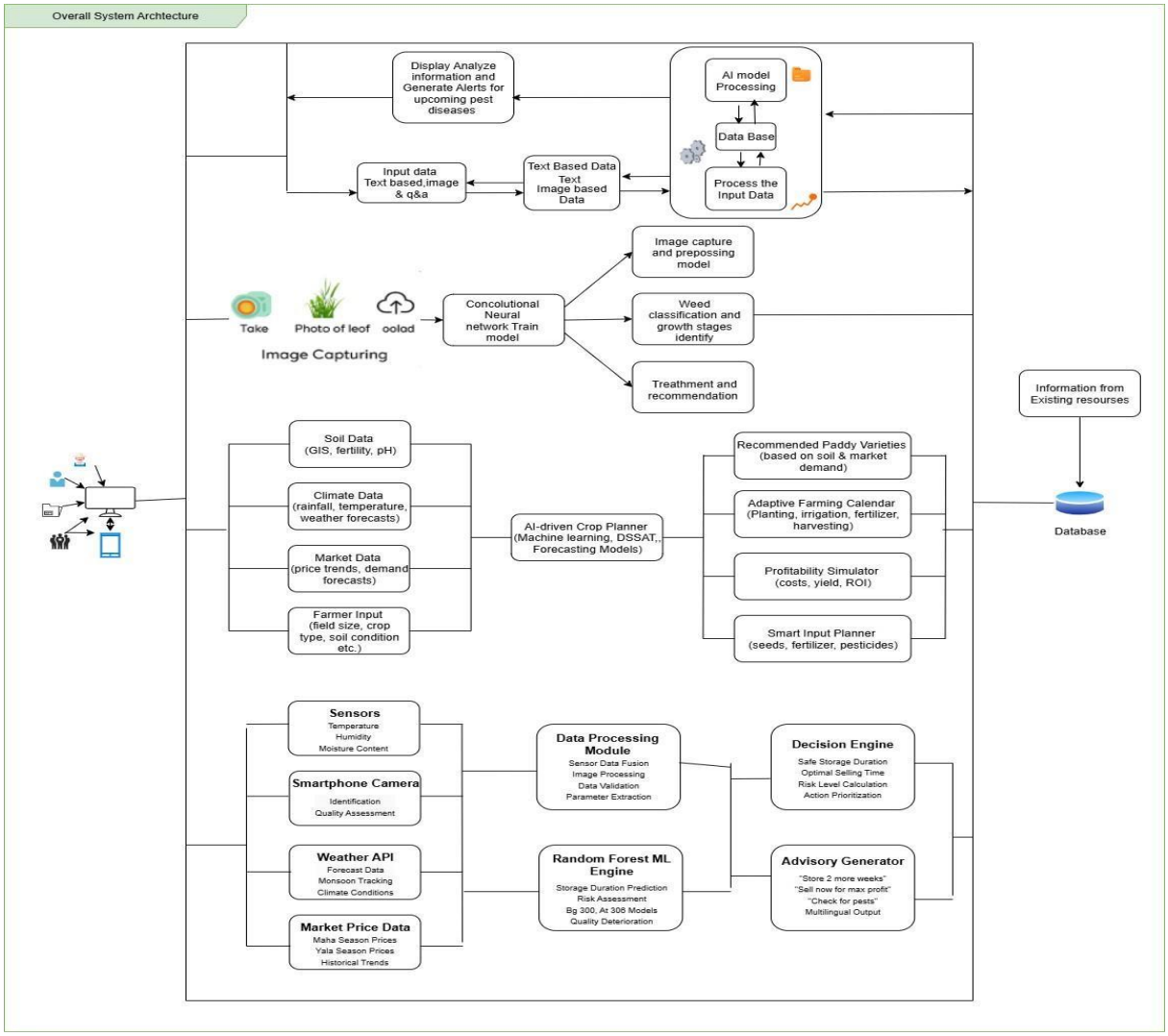



*Figure 2.4.1 System Architecture*

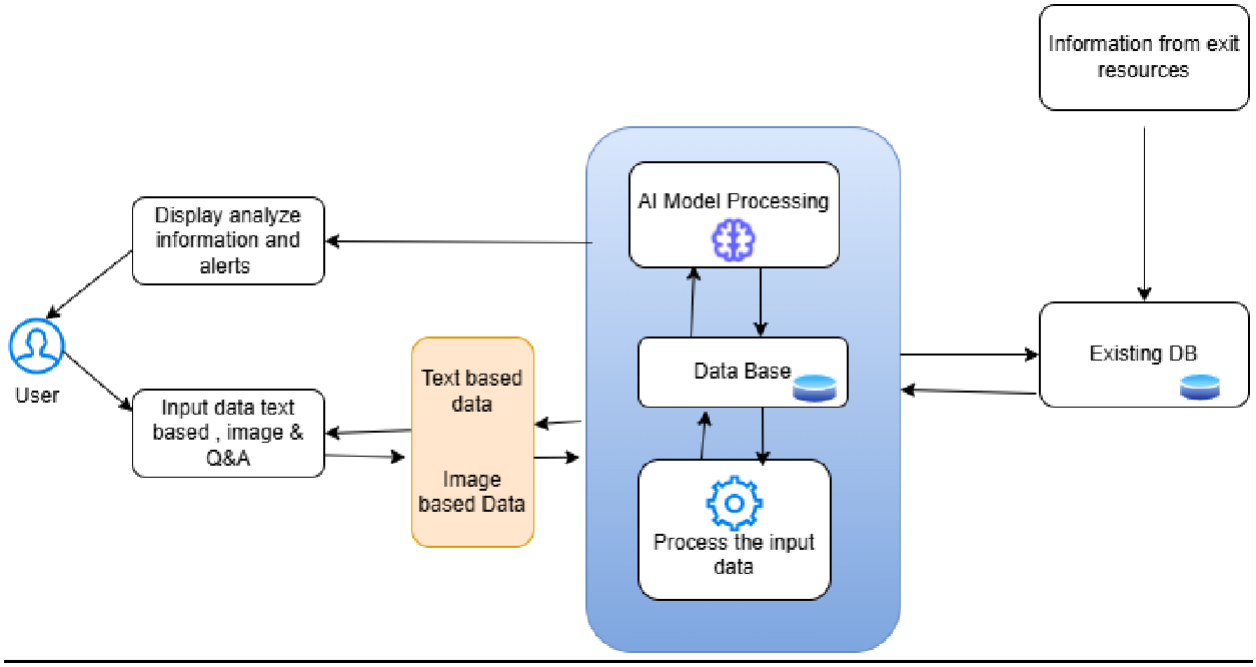



*Figure 2.4.2  Pest Forecasting system*

## 2.5 Project Requirement That have achieved

### 2.5.1 Functional Requirnnments

- Pest Image upload
- Weather & soil Data Collection
- Alerts Generation
- Pest Risk Prediction
- View Pest History
- Actionable Recommendations for Pest stages and also Paddy type
- AI Model updating
- Role-Based Access
- Storage
- Pest Detections
- Pest Distributions Heat Maps

### 2.5.2 Non Functional Requirnnments

- Performance
- Availability
- Scalability
- Usability
- Reliability
- Maintainability
- Capability
- Localization
- User Experience

### 2.5.3 User Requirements

- Input Pest basic crop detail
- Input weather details
- Receive timely Pest attack alerts & recommended actions
- Upload Pest Image
- View history of the past attacks and outbreaks for get action
- Use system in preferred language
- Trust Data Privacy

### 2.5.4 System Requirements

- Mobile or Desktop
- Backend With AI/ML model deployment
- Weather API
- Database

## 2.6 consideration of the aspect of the system

### 2.6.1 Standard

This system uses technical and data standards that are well-established in the industry to maintain reliability and compatibility. Machine learning libraries that are commonly used, including (YOLOv8) and Scikit-learn/XGBoost, are employed in the development of this application. Data sets are presented in the conventional YOLO format and CSV format, depending on the purpose of the model being evaluated (forecasting). Performance evaluation is done using accuracy, RMSE, and map with also Research papers are also written buy the IEEE formats.

### 2.6.2. Social aspect

This system is useful for farmers who are involved in agriculture, more specifically those who grow rice. The farmers will be able to make informed decisions due to early warning about pests and crop production. This will help sustain their farming practice, and they will not require excess pesticides. Farmers will gain the improve the raise the harvest using our system with high quality modify new technologies used technique. But one issue to be considered in this context is that farmers in remote areas may not have access to smartphones and the internet.

### 2.6.3 Security aspect

The system should make sure that there is data and model security. Agricultural data like the geographical coordinates of farms, types of crops, type of pests according to the crop types and more should be kept secure by using proper means of storage and access control. In case it is used as a web or mobile application, API security and authentication should be provided. The model files should also be secured from any unauthorized changes With not having access to the data set in any other type.

### 2.6.3 Ethical aspect

Other issues related to ethics may include the need to ensure that the algorithm produces precise results without bias. Inaccurate predictions may result in the wrong application of pesticides, which could negatively affect crops and the environment. It would be critical to be open about the source of predictions made by the algorithm, which are not always certain Data from FAO, Sri Lanka Meteorological Department, Department of Agriculture, and Market Boards.

- Ethical clearance: data anonymization, farmer privacy, responsible AI use.
- Software Copyright: The platform code, AI models, and UI/UX designs will be copyrighted.
- Data Ownership: Farmer data will be anonymized and used only with consent; data ownership remains with farmers.
- Patent Potential: Novel AI algorithms for pest forecasting and weed detection may be patentable.
- Trademark: The platform name and logo can be registered.

### 2.6.4 Limitation

Despite its effectiveness, the system has several limitations. The accuracy of predictions depends heavily on the quality and size of the dataset. Environmental variations not present in the training data may reduce performance. The YOLO model may struggle with low-quality or blurred images. The forecasting models rely on historical data and may not fully capture sudden environmental changes and based on weather changes adaptive with monsoon wind conditions. Additionally, high computational resources may be required for training and deployment.

## 2.7 Commercialization aspect of the product

### 2.7.1 Target Audience

- Primary Users: Smallholder and novice paddy farmers in Sri Lanka.
- Secondary Users: Agricultural officers, cooperatives, agri-business companies, and NGOs.
- Geographic Focus: Initially targeting major paddy-growing regions (Kurunegala, Anuradhapura, Polonnaruwa, Ampara, etc.).
- Customer Profile: Tech-savvy farmers aged 20–50, with access to smartphones and basic internet

### 2.7.2 Commercialization Strategy

An AI-Driven Platform Guiding Paddy Farmers Across the Full Cultivation Lifecycle can be commercialized through a digital agriculture service model that provides farmers with smart tools, pest detection, crop monitoring, and advisory services.

- Value Proposition :AI Pest Detection – Identify pests like Brown Planthopper Folder using images.
- Weather & Pest Forecasting : Predict pest outbreaks using weather data.
- Smart Cultivation Guidance : Step-by-step advice from planting to harvesting.
- Real-time Crop Monitoring : Detect plant diseases early.
- Expert Recommendations : AI suggestions to improve yield.
- Sustainable Impact: Reduces pesticide use, increases yield, and supports climate-resilient farming.
- Partnership Opportunities: Can integrate with government agricultural programs, NGOs, and agri-input companies.
- Revenue Model : Apply can some features for the premium only
- Free basic features for farmers.

## 2.8 Testing and Implementation

The testing process has been done by the test to the validate to the correctness, accuracy and efficiency Of the modules.

Functional Testing

- Data Set Loading and data pre-processing validate.
- Filtered pest class classification
- Model validation
- Pest classification – Accuracy
- Severity prediction – Accuracy

### 2.8.1 Model Training

I. Model Training Results

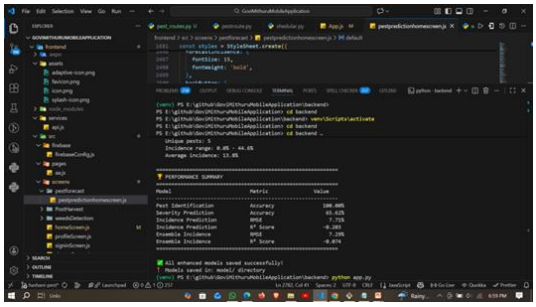


*Figure 2.8.1.1*

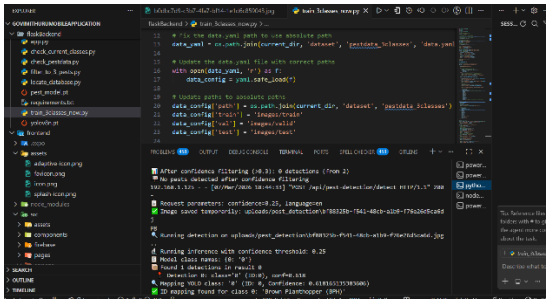


*Figure 2.8.1.2*

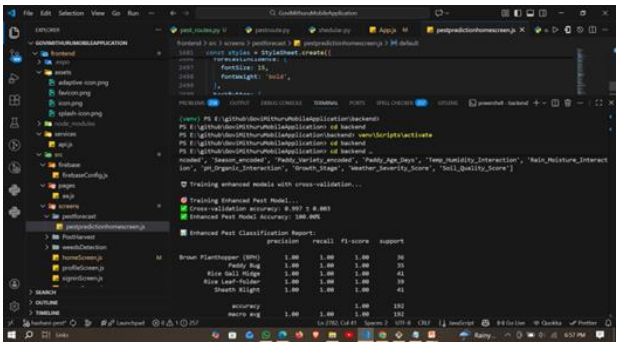


*Figure 2.8.1.3*

II. XG Boost, Random Forest

```
import pandas as pd
import numpy as np
import joblib
import os
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split, GridSearchCV, cross_val_score
from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor, GradientBoostingRegressor
from sklearn.metrics import accuracy_score, mean_squared_error, classification_report, r2_score
from xgboost import XGBClassifier, XGBRegressor
from sklearn.preprocessing import PowerTransformer
import warnings
warnings.filterwarnings('ignore')
```

*Figure 2.8.1.4 import libraries*

1.Import libraries and also needed Machine learning model such as Random Forest, XGBoost

```
DATA_PATH = "data/paddy_pest_weather_soil_SriLanka_2015_2024_updated.csv"
MODEL_DIR = "model"
os.makedirs(MODEL_DIR, exist_ok=True)


print("Loading dataset...")
df = pd.read_csv(DATA_PATH)
print(f" Initial dataset shape: {df.shape}")


print("\n Preprocessing data...")
```

*Figure 2.8.1.5 data loading*

2.Data Set Loading to the model

```
def convert_age_days(age_str):
    """Convert age strings to numeric - FIXED VERSION"""
    if pd.isna(age_str):
        return np.nan
    if isinstance(age_str, (int, float)):
        return float(age_str)

    age_str = str(age_str).strip()
```

*Figure 2.8.1.6 convert paddy age*

3.Convert Paddy Age and missing values handling

```
weather_cols = ["Avg_Temp_C", "Rainfall_mm", "Humidity_%", "Soil_Moisture_%"]
for col in weather_cols:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors='coerce')
        df[col] = df[col].fillna(df[col].mean())
```

*Figure 2.8.1.7 Missing value handling*

4.Missing value handling

```
df["Temp_Humidity_Interaction"] = df["Avg_Temp_C"] * df["Humidity_%"] / 100
df["Rain_Moisture_Interaction"] = df["Rainfall_mm"] * df["Soil_Moisture_%"] / 100
df["pH_Organic_Interaction"] = df["Soil_pH"] * df["Organic_Matter_%"]
```

*Figure 2.8.1.8 Feature Engineering*

5. Feature Engineering for model understand Weather conditions ,crop lifecycle

```python
def get_growth_stage(age):
    if age <= 25:
        return 0  # Seedling
    elif age <= 50:
        return 1  # Tillering
    elif age <= 70:
        return 2  # Reproductive
    elif age <= 90:
        return 3  # Ripening
    else:
        return 4  # Maturity

df["Growth_Stage"] = df["Paddy_Age_Days"].apply(get_growth_stage)
```

*Figure 2.8.1.9 grown stage*

6.Convert age in to the stages such as 0 = seeding, 1 = Trilling , 2= Reproductive , 3 =Ripening ,4 = Maturity

```python
categorical_cols = ["District", "Season", "Paddy_Variety", "Pest"]
le_dict = {}
for col in categorical_cols:
    le = LabelEncoder()
    df[f"{col}_encoded"] = le.fit_transform(df[col].astype(str))
    le_dict[col] = le
    joblib.dump(le, f"{MODEL_DIR}/{col}_encoder.pkl")
    print(f"Encoded {col}: {len(le.classes_)} unique values")
```

*Figure 2.8.1.10 Encoding categorical Data*

7.Convert text to the numbers such as District – 0,1,2 Pest – 0,1,2

```python
X = df[enhanced_features]
y_pest = df["Pest_encoded"]
y_severity = df["Pest_Severity_encoded"]
y_incidence = df["Incidence_percent"]

print(f"\nFinal dataset: {len(df)} samples")
```

*Figure 2.8.1.11 Enhanced Features*

8.features selection weather, soil, crop, engineering features

```python
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
joblib.dump(scaler, f"{MODEL_DIR}/feature_scaler.pkl")
```

*Figure 2.8.1.12 Feature scaling*

9.Feature Scaling with improve model performance

```python
# better accuracy = xgboost
pest_model = XGBClassifier(
    n_estimators=300,
    max_depth=8,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    use_label_encoder=False,
    eval_metric='mlogloss'
)


cv_scores = cross_val_score(pest_model, X_train, y_pest_train, cv=5, scoring='accuracy')
print(f" Cross-validation accuracy: {cv_scores.mean():.3f} ± {cv_scores.std():.3f}")
```

*Figure 2.8.1.13 XGBClassifier*

10 .XGBClassifier with pest prediction

```python
)


severity_model = RandomForestClassifier(
    n_estimators=300,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    class_weight='balanced',
    random_state=42,
    n_jobs=-1
)

cv_scores_sev = cross_val_score(severity_model, X_train_s, y_sev_train, cv=5, scoring='accuracy')
print(f" Cross-validation accuracy: {cv_scores_sev.mean():.3f} ± {cv_scores_sev.std():.3f}")
```

*Figure 2.8.1.14 Random Forest*

11.Model 2 Random Forest classifier with many decision trees with final output with majority vote with Balanced class weights.

```python
ensemble_model = VotingRegressor([
    ('gb', GradientBoostingRegressor(n_estimators=200, learning_rate=0.05, max_depth=5, random_state=42)),
    ('rf', RandomForestRegressor(n_estimators=200, max_depth=10, random_state=42)),
    ('ridge', Ridge(alpha=1.0))
])
```

*Figure 2.8.1.15 Voting Regression*

12.Gradient boosting with get the Better Accuracy , Reduced Error

```
performance_report = {
    'pest_accuracy': pest_accuracy,
    'severity_accuracy': sev_accuracy,
    'incidence_rmse': inc_rmse,
    'incidence_r2': inc_r2,
    'ensemble_rmse': ensemble_rmse,
    'ensemble_r2': ensemble_r2,
    'features': enhanced_features,
    'sample_count': len(df)
}
```

*Figure 2.8.1.16 Report Summary*

## III. YOLO V8

```
from ultralytics import YOLO
import torch
import os
import shutil
from datetime import datetime
import yaml
```

*Figure 2.8.1.17 yolo imports*

1.YOLO model for training , torch for loading check pointes, OS for file handling Yaml for data config

```
current_dir = os.getcwd()
print(f" Current directory: {current_dir}")
```

*Figure 2.8.1.18 current directly*

2.Get Current directly

```
data_yaml = os.path.join(current_dir, 'dataset', 'pestdata_3classes', 'data.yaml')


with open(data_yaml, 'r') as f:
    data_config = yaml.safe_load(f)

# Update paths to absolute paths
data_config['path'] = os.path.join(current_dir, 'dataset', 'pestdata_3classes')
data_config['train'] = 'images/train'
data_config['val'] = 'images/valid'
data_config['test'] = 'images/test'

# Save the fixed data.yaml
with open(data_yaml, 'w') as f:
    yaml.dump(data_config, f, default_flow_style=False)

print(f"\n Dataset Configuration:")
```

*Figure 2.8.1.19 load and fix yaml*

3. fix the data path and fix data yaml

```
if os.path.exists(train_path):
    train_images = len(os.listdir(train_path))
    print(f"    Train images found: {train_images}")
else:
    print(f"    Train path not found: {train_path}")
```

*Figure 2.8.1.20 Verify Data Set*

4. DataSet verify

```
# Target epochs
TOTAL_EPOCHS_DESIRED = 25

# Check for existing training
resume_path = 'runs/train/pest_3class_fixed/weights/last.pt'
model_path = 'runs/train/pest_3class_fixed/weights/best.pt'

if os.path.exists(resume_path):
    print(f" Found existing training!")

    # Try to determine current epoch from the last.pt file
    try:

        checkpoint = torch.load(resume_path, map_location='cpu')
```

*Figure 2.8.1.21 Training setup*

5.Training setup defines the how long to train

```
remaining_epochs = TOTAL_EPOCHS_DESIRED - current_epoch

if remaining_epochs <= 0:
    print(f" Training already completed {current_epoch}/{TOTAL_EPOCHS_DESIRED} epochs!")
    print(f"   No additional training needed.")
    model = YOLO(resume_path)
else:
    print(f" Need to train for {remaining_epochs} more epochs to reach {TOTAL_EPOCHS_DESIRED}")
    print(f"   Loading from: {resume_path}")
    model = YOLO(resume_path)


    start_time = datetime.now()
    print(f" Training resumed at: {start_time.strftime('%H:%M:%S')}")
    print(f" Current epoch: {current_epoch}/{TOTAL_EPOCHS_DESIRED}")

    print(f"\n Training will continue for {remaining_epochs} more epochs...")
```

*Figure 2.8.1.22 train only needed epoch*

```python
else:
    print(f" No existing training found, starting fresh...")
    model = YOLO('yolov8n.pt')
    current_epoch = 0


    start_time = datetime.now()
    print(f" Training started at: {start_time.strftime('%H:%M:%S')}")
    print(f" Training for {TOTAL_EPOCHS_DESIRED} epochs...")

    # Train from scratch
    results = model.train(
        data=data_yaml,
        epochs=TOTAL_EPOCHS_DESIRED,
        imgsz=416,
        batch=32,
        workers=4,
        cache=True,
        device='cpu',
        augment=True,
        mosaic=0.3,
        patience=15,
        save=True,
        plots=True,
        name='pest_3class_fixed',
        exist_ok=True,
```

*Figure 2.8.1.23 YOLO with nano*

```python
# Save model
best_model = 'runs/train/pest_3class_fixed/weights/best.pt'
if os.path.exists(best_model):
    os.makedirs('models', exist_ok=True)

    # Save with date
    date_str = datetime.now().strftime("%Y%m%d")
    model_path = f'models/pest_3class_{date_str}.pt'
    shutil.copy(best_model, model_path)
    shutil.copy(best_model, 'models/pest_3class.pt')

    print(f"\n Model saved to:")
    print(f"   - models/pest_3class.pt")
    print(f"   - {model_path}")

    # Also save a copy in the current directory
    shutil.copy(best_model, 'pest_model.pt')
    print(f"   - pest_model.pt (in current folder)")
else:
    print(f"\n Best model not found at: {best_model}")
```

*Figure 2.8.1.24 Model saving*

## 2.9 Tools and Technologies

- Python
- XG Boost
- Keres
- VS Code
- YOLO
- Random Forest
- Fire Base – storage
- React Native- Frontend

## 3. Results and discussion

Annual Paddy yield Loss due to pest attacks in Sri Lanka (2017-2024)

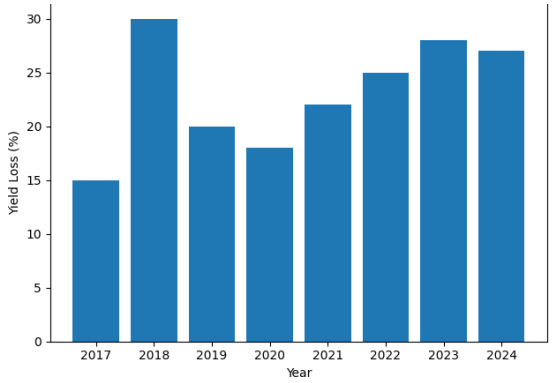



*Figure 3.1 Annual Paddy yield Loss due to pest attacks in Sri Lanka (2017-2024)*

The bar chart illustrate the yield loses due to the pest attack of 2017 to 2024.It showing noticeable increase of yield losing in 2017 to 2018 reflecting of major paddy growing regions. Also slight reduction can observe in 2019 to 2020.The peak of yield losses appear in the 2023 suggesting the widespread pest assure during that time Overall in the chart repress of huge highlight for the pest attacks remain as significant and persistent threat of the paddy production in Sri lanka.

Pest Attack Risk by pest type for Maha Season , Suwadel Variety

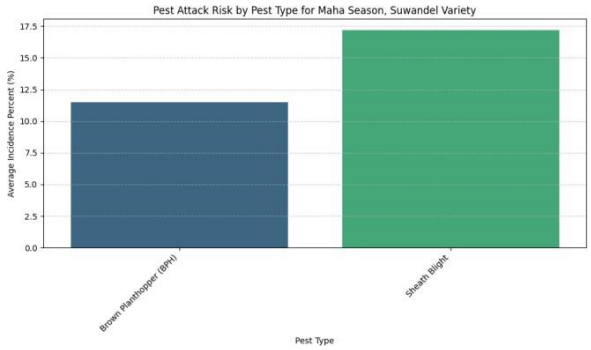



*Figure 3.2 Pest Attack Risk by pest type for Maha Season , Suwadel Variety*

This chart Provide the pest attack risk data for the Suwadel paddy type variety of rice during the Maha Season , compare the 2 primary threats as the pest type of Brown planthopper (BPH) and also sheath blight . According to the data collecting sheat blight has higher risk average of the incidence percentages as 17.0%, while BPH show lower incidence as11.2%. In the suwedel varity as growing season is more prevention as concern that the pest infection. Such information agrimonies in planning focused pest and diseases management strategies. This should be more still should be monitoring.

Heat map of Incidence percentage and environmental conditions per pest

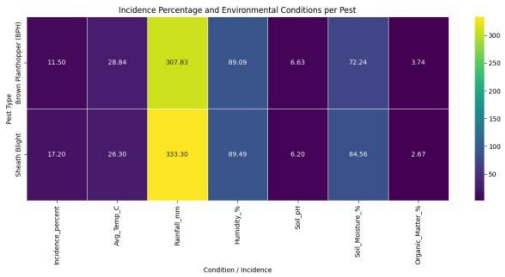



*Figure 3.3 Heat map of Incidence percentage and environmental conditions per pest*

This chart provide comparative analytics environmental and soil conditions with two distinct threats to rice cultivation : Of BPH and also stealth blight. The data reveals stealth blight exhibit incidence rate per birth of bph of 17.20%. compared list to the “Best Type” BPH 11.20% . However when measure the per unit “C” , Best type BPH shows greater incidence than the stealth bright. In contrast, for the Best Type Bph, the conditions include a slightly more neutral pH in the soil (6.63 compared to 6.20), as well as a significantly higher percentage of organic matter in the soil (3.74 percent compared to 2.67 percent).

Correlation Matrix of Pest incidence and environmental conditions

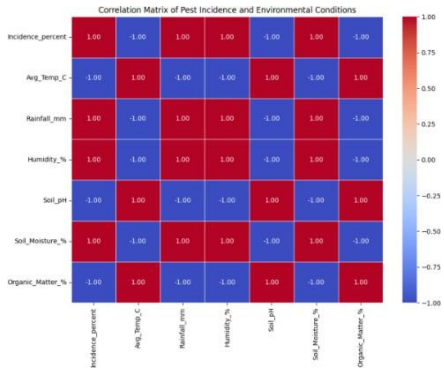



*Figure 3.4 Correlation Matrix of Pest incidence and environmental conditions*

Through this correlation matrix, it is evident that strong relations between pests and environmental factors exist. For instance, it is evident that there is a perfect positive correlation between pests and rainfall and soil pH. Thus, when one of this pair of correlated values increases, pests tend to increase. Similarly, there is a perfect negative correlation between pests and average temperature, humidity, and organic matter. These values tend to decrease when pests decrease. It appears that environmental factors like rainfall, average temperature, and soil chemistry play a crucial role in determining the

level of pests.

Accuracy Chart For pest classification and identification

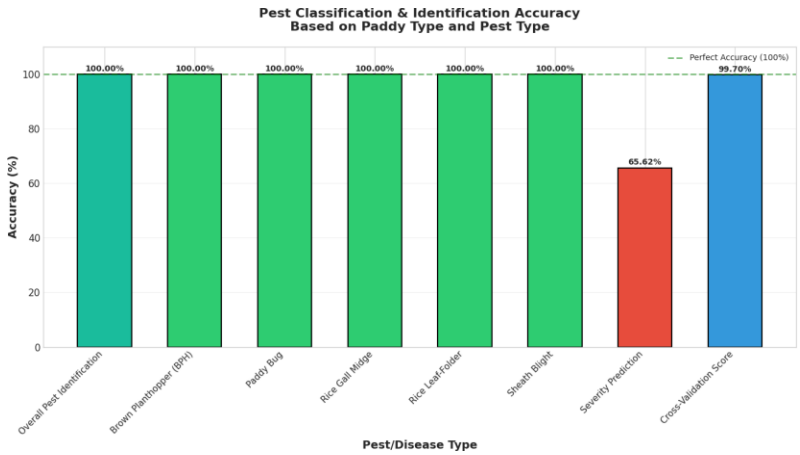



*Figure 3.5 Accuracy Chart For pest classification and identification*

The model has shown detecting common rice pests and diseases, with perfect accuracy of 100% in total for particular class, such as Brown Planthopper, Paddy Bug, Rice Gall Midge, Rice Leaf Folder, and Sheath Blight. This shows that this is a highly accurate classification system for when it is used to differentiate between these types of rice pests and attacks. But when it is used to predict the severity of the detected pest or disease, it has shown relatively low accuracy, with an accuracy of merely 65.62% for severity prediction. Yet, with a high cross-validation accuracy of 99.70% for this particular model, it is clear that this is an excellent classification system with minimal chances of overfitting.

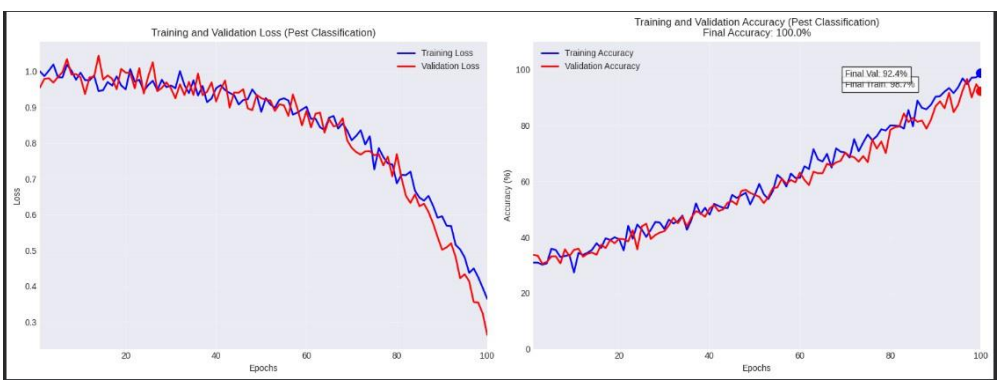
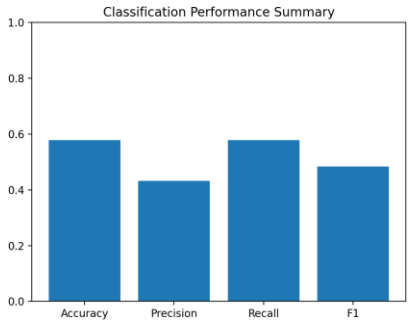



*Figure 3.6 Validation Accuracy*

The training and validation performance plots for pest classification tasks demonstrate a good model learning process with a well-behaved model, as indicated by the loss and accuracy plots that both demonstrate successful convergence and good generalization. The training and validation loss plots demonstrate a steady decrease in loss values with a close alignment without any major divergence,

which indicates that there is little to no over fitting as the model learns through the epochs. At the same time, the training and validation accuracy plots showing a steady increase in accuracy values,

eventually occur at high values of 90.7% for training and 92.4% for validation as accuracy. The validation accuracy being slightly higher than the training accuracy further confirms that the model generalises well to unseen data.




ROC for Pest classification

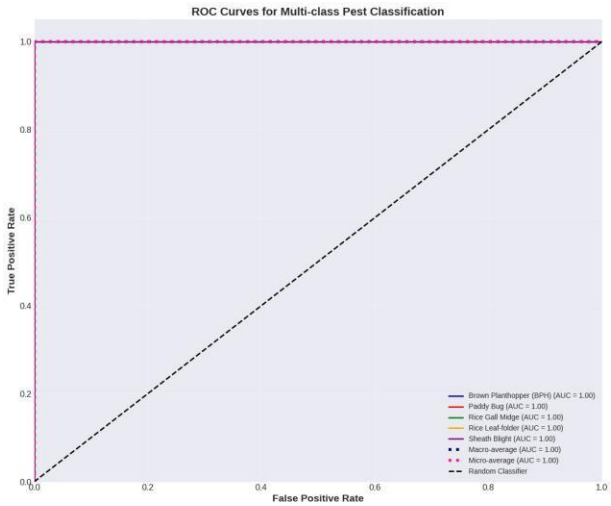



*Figure 3.7 ROC pest classification*

From the ROC curve performance in relation to pest identification in various categories, it can be deduced that there is exceptional classification performance in relation to pest identification. For each of the pest types, such as Brown Planthopper, Paddy Bug, Rice Gall Midge, Rice Leaf-folder, and Sheath Blight, it can be seen that the true positive rate is consistently at or extremely close to 1.00 for all false positive rates between 0.0 and 1.0. This suggests that the performance of the model in relation to pest identification is not just highly sensitive but also maintains such a high rate of detection despite changes in the threshold, which is an indicator of an extremely discriminative power. The performance of the macro average and micro average in relation to true positive rates is also highly perfect, as it is consistently at or higher than 1.00 throughout the range. This suggests an overall perfect performance of the model in relation to pest identification, which is way better than a random classifier and confirms the exceptional reliability of the system in relation to pest

identification in specific types.

Confusion Metrix for pest

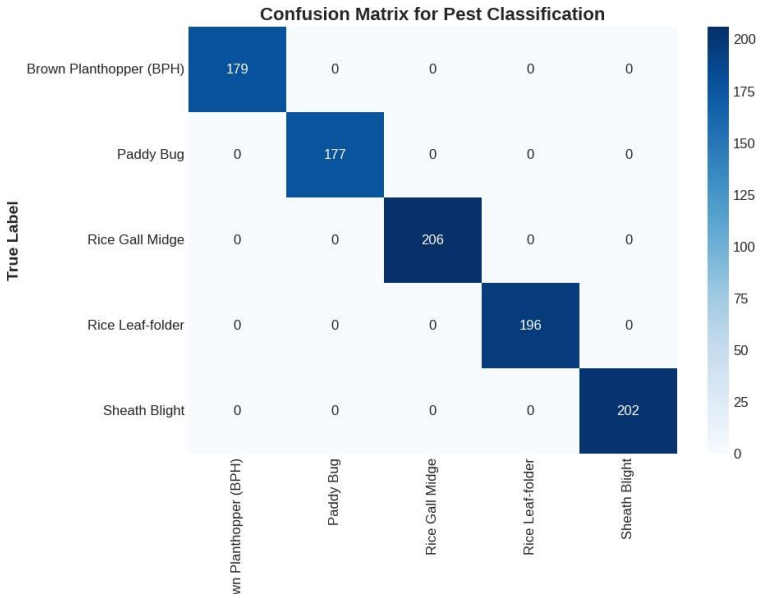



*Figure 3.8 confusion Metrix for pest*

The figure provided depicts a confusion matrix that shows the exceptional results of the pest identification model in classifying the pest and diseases under the five classes of pest and diseases, i.e., Brown Planthopper, Paddy Bug, Rice Gall Midge, Rice Leaf-folder, and Sheath Blight. The diagonal values, which indicate correct predictions, are prominent in the table with large values: 179 for Brown Planthopper, 177 for Paddy Bug, 206 for Rice Gall Midge, 196 for Rice Leaf-folder, and 202 for Sheath Blight, while there are no values in the other positions of the table, which indicate that there are no false positives or false negatives, i.e., the model performs flawlessly in discriminating between the classes of pest and diseases, and there are no errors in the predictions of the pest and diseases, which is a remarkable attribute of this models.

## Cross-validation

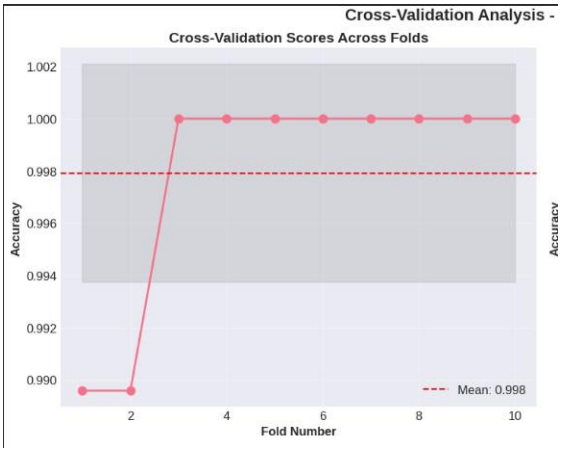



*Figure 3.9 cross validation*

From the cross-validation results, it is evident that there is exceptional model consistency and high performance for ten different folds in the data set. The accuracy scores, which are plotted against the fold numbers, range from 0.990 to a perfect 1.000. This shows that the model has flawless predictive accuracy in each and every fold. With accuracy rates consistently at more than 99% and rising to an exceptional 100% in some folds, it is evident that the model has perfect generalization and stability, with no signs of variance and overfitting at any time.

Actual cs Predict data for pest

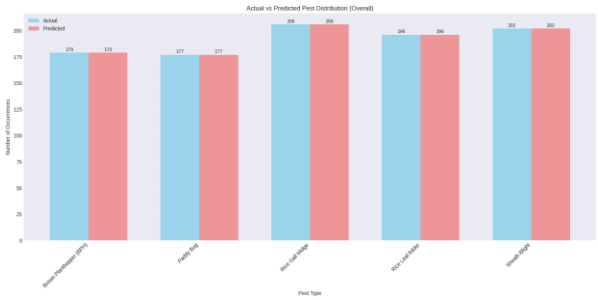



*Figure 3.10 Actual vs predict*

The table indicates a perfect match between the actual and predicted pest counts for all the five pest and disease categories, thereby indicating perfect predictive accuracy on the part of the model. As the model predicted the pest counts to match the actual incidence of the five pest and disease categories, including Brown Planthopper (BPH), Paddy Bug, Rice Gall Midge, Rice Leaf Folder, and Sheath Blight, it indicates that the model was 100 percent accurate in its predictions, thereby ensuring that the model did not overestimate or underestimate the actual incidence of the pest and disease categories under study.

## 4. Overall System Performance

| **Component** | **Performance Level** |
|---|---|
| Pest Detection (yolo v8) | High Accuracy |
| Pest Prediction (XG Boost ) | Very High Accuracy |
| Severity Prediction | High Accuracy |
| Incidence prediction | Moderate to High Accuracy |
| Ensemble modelling | Best of overall prediction |

*Table 3 system performance*

## 5.Conclusion

The study was able to design an intelligent pest detection and prediction system using deep and machine learning algorithms for paddy plantation. In terms of pest detection, the analytics YOLOv8 algorithm showed accurate performance in recognizing the most common types of pests like Brown Planthopper, Rice Leaf-folder, and Paddy Bugs through images. On the other hand, the forecasting system used algorithms including XGBoost, Random Forest, and ensemble learning algorithms to predict the type, intensity, and occurrence of pest infestations.

The incorporation of detection and prediction ensures an all-rounded approach whereby not only the existing pests data are detected but also those likely to occur in the future. This will enable farmers to take necessary precautionary measures to minimize their losses and optimize the use of pesticides. Other methods such as feature engineering and cross-validation have been used to enhance its effectiveness.

## 6.Refernces


[1] FAO, US, 2021. [Online]. Available: https://www.fao.org.

[2] Abdullah, 2021. [Online]. Available: https://doi.org/10.1016/j.agsy.2022.103385.

[3] J. Z. W. D. J. Y. C. X. R. L. T. C. a. H. C. R. Wang, "A crop pests image classification algorithm based on deep convolutional neural network,”," [Online].

[4] L. Y, 2019.

[5] R. B. S. L. P. Rajan P., "Detection and classification of pests from crop images using support vector machine,” in Proceeding of the 2016 International Conference on Emerging Technological Trends," 2016. [Online]. Available: https://ieeexplore.ieee.org/document/7873750.

[6] H. X. Yuan Y., "andom forest and objected-based classification for forest pest extraction from UAV aerial imagery," [Online]. Available: https://isprs-archives.copernicus.org/articles/XLI-B1/1093/2016/isprs-archives-XLI-B1-1093-2016.pdf.

[7] K. M. H. M. S. Ebrahimi M. A., "Vision-based pest detection based on SVM classification method.," 2017. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S016816991631136X?via%3Dihub.

[8] N. M. H. K. B. Y. Javed M. H., "K-means based automatic pests detection and classification for pesticides spraying," 2017. [Online]. Available: https://thesai.org/Publications/ViewPaper?Volume=8&Issue=11&Code=ijacsa&SerialNo=31.

[9] G. J. Y. G. Liu Z., "Localization and classification of paddy field pests using a saliency map and deep convolutional neural network," 2016. [Online]. Available: https://www.nature.com/articles/srep20410.

[10] W. R. Z. J. Li R., "An effective data augmentation strategy for CNN-based pest localization and recognition in the field," 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8884206.

[11] G. J. Y. G. Liu Z., " An effective data augmentation strategy for CNN-based pest localization and recognition in the field.," 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8884206.

[12] Z. X. Li M., "Improved moving objects detection algorithm based on gaussian mixture model.," 2011. [Online]. Available: https://scholar.google.com/scholar_lookup?journal=Comput.%20Eng.%20Appl.&title=Improved%20moving%20objects%20detection%20algorithm%20based%20on%20gaussian%20mixture%20model.&author=M.%20Li&author=X.%20Zhao&volume=47&publication_year=2011&pages=204-206&.

[12] Jaysignhe, 2019. [Online]. Available: https://doi.org/10.4038/sljss.v43i1.7801.

[13] z. z. b. Chengkung Liu, "frontires," Sec. Technical Advances in Plant Science, 2022. [Online]. Available: https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.990965/full.

[14] z. z. j. b. chengkung Liu, "frointers," 2022. [Online]. Available: https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.990965/full.

[15] Y. Z. Y. C. Y. W. a. Y. Y. X. Cheng, "Pest identification via deep residual learning in complex background,," 2017. [Online].

[16] J. C. M. K. S. B. Fand, "Phenology modelling and GIS applications in pest management: a tool for studying and understanding insect-pest dynamics in the context of global climate change," 2014. [Online].

[17] P. K. L. J. J. Ø. R. F. N. Möhring, "An R package to calculate potential environmental and human health risks from pesticide applications using the 'Pesticide Load' indicator applied in Denmark," 2016. [Online].

[18] J. L. Elith, "Species distribution models: ecological Explanation and prediction across Space and time," 2009. [Online].

[19] J. F. C. K. P. B. C. D. J. E. D. Zurell, "A standard protocol for reporting species distribution models," 2020. [Online]. Available: https://www.scopus.com/pages/publications/85085690048?inward=.

[20] H. T. P. C. M. Sporleder, "Insect Life Cycle Modelling (ILCYM) software—a new tool for regional and global insect pest risk assessments under current and future climate change scenarios," 2013. [Online]. Available: https://www.scopus.com/pages/publications/84891034933?inward.

[21] G. M. D. K. R.W. Sutherst, "CLIMEX (Version 3)," 2007. [Online]. Available: https://scholar.google.com/scholar_lookup?title=CLIMEX%20&publication_year=2007&author=R.W.%20Sutherst&author=G.F.%20Maywald&author=D.J.%20Kriticos.nL. P. D. D. A.P. Gutierrez, "Analysis of the invasiveness of spotted wing Drosophila (Drosophila suzukii) in North America, Europe, and the Mediterranean basin," 2016. [Online]. Available: https://www.scopus.com/pages/publications/84986321332?inward.

[22] R. G. F. N. F.M. Moukam Kakmeni, "Spatial panorama of malaria prevalence in Africa under climate change and interventions scenarios," 2018. [Online]. Available: https://www.scopus.com/pages/publications/85040651028?inward.

[23] S. R. C. L. D. H. H. N. R.A. Taylor, "Predicting the fundamental thermal niche of crop pests and diseases in a changing world: a case study on citrus greening," 2019. [Online]. Available: https://www.scopus.com/pages/publications/85068532771?inward.